LEARNMATE

Scalability & Future Plan

Field	Details
Date	30/09/2026
Team ID	SWTID-2026-2032
Project Name	Learnmate – AI StudyBuddy
Maximum Marks	1 Mark (as stated in the supplied template)

Scalability & Future Plan

This document outlines how the current Learnmate solution, based on the supplied backend code screenshots, could be extended to support a larger user base, increased data loads, and additional study-support features. It also presents a proposed roadmap beyond the current implementation. Proposed upgrades are future recommendations, not features confirmed as already implemented.

Source boundary: The screenshots show a Node.js/Express backend, MongoDB with Mongoose, JWT-based authentication, Multer uploads, and a Gemini utility. They do not provide load-test results, deployment architecture, current user volume, or operational metrics. The current-state descriptions below are therefore limited to visible code evidence.

Current System Limitations

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	No evidence of load balancing, horizontal scaling, or load-test results in the supplied code screenshots.	The capacity for handling concurrent users and peak traffic has not been established.	High
2	The visible Material model stores content and learning fields in MongoDB; no storage strategy, retention policy, or large-file storage service is shown.	Growth in uploaded materials and generated content may increase database and disk usage; backup and retention behavior is not established.	Medium
3	The visible upload middleware stores files locally using Multer disk storage and sets a 5 MB file-size limit.	Local disk capacity and file availability across multiple server instances may become constraints as usage grows.	High

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Express server starts as a single application process in the visible entry file; no load-balancing setup is shown.	Deploy multiple stateless API instances behind a load balancer; use measured traffic and load tests to size instances and define scaling thresholds.
2	Data Storage	MongoDB/Mongoose is used. Material documents include content, summary, flashcards, quiz and studyPlan. Uploaded files are stored on local disk in the visible Multer setup.	Use managed MongoDB with backup and monitoring; consider object storage for uploaded files and store file references/metadata in MongoDB. Define retention and indexing based on actual query patterns.
3	Performance	No caching, queue, background-job system, or performance metrics are visible in the supplied screenshots.	Profile API and AI request latency; add caching for suitable repeated reads and a background job queue for longer AI processing. Apply pagination and database indexes where supported by usage patterns.

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
4	Security	Visible controls include password hashing with bcryptjs, JWT verification, student/admin roles, upload extension filtering, and a 5 MB upload limit. No complete security audit is provided.	Use HTTPS, secure secret management, rate limiting, stricter input validation, authorization tests, dependency updates, and centralized logging/monitoring. Review upload content validation and access controls before deployment.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Stabilize the current MVP: verify authentication and role-protected routes, test material workflows and supported uploads, validate Gemini integration, and add basic API tests and error logging.	Short term (0–2 months)	Improves reliability and provides a tested baseline before scaling.
Phase 2	Improve data and file handling: add database indexes based on query patterns, backups, upload storage planning, and pagination for material lists.	2–4 months	Supports increasing material volume and improves data management.
Phase 3	Introduce performance and scale improvements: run load tests, measure API/AI latency, add background processing for longer AI tasks, and deploy multiple API instances if required.	4–6 months	Helps the system handle higher concurrent usage and reduces delays for long-running tasks.
Phase 4	Extend study-support capabilities and operational readiness: refine summaries, flashcards, quizzes and study plans; add monitoring, security review, and user feedback-driven improvements.	6–12 months	Expands learning support while improving maintainability, security and user experience.

Conclusion

The visible Learnmate backend provides a starting point for future growth through modular API routes, MongoDB models, authentication middleware, upload handling, and a Gemini integration. Before scaling, the team should verify end-to-end behavior and measure actual usage and performance. The roadmap prioritizes a tested MVP, reliable data and file handling, measured performance improvements, and then broader study-support and operational enhancements.

Prepared using the supplied Learnmate code screenshots and the uploaded “Scalability & Future Plan” template. Dates, priorities and roadmap timelines are proposed and should be confirmed by the project team before submission.